

USING

SEMANTIC LINK

TO GET AN OVERVIEW OF POWER BI
IN THE LANDS OF MIDDLE EARTH

SESSION TARGET



- SKILLED python coder
- EXPECTING deep dive
- BEST practices



- NEVER WRITTEN python
- FIRST STEPS INTO FABRIC MONITORING



HOBBITS

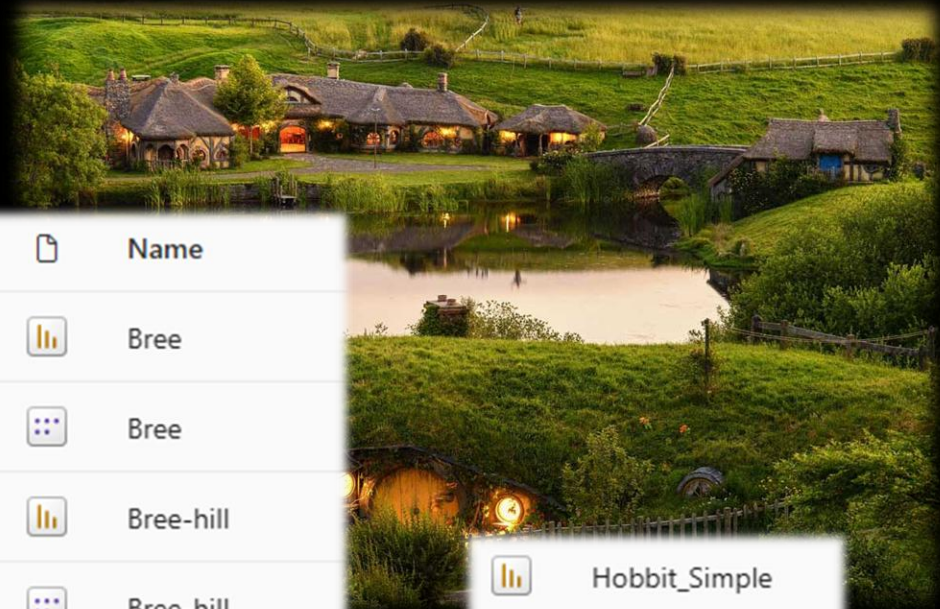
SMALL TEAMS

KNOW EACH OTHER

TRADITION

NO FORMAL AGREEMENT

INNOVATION BY INDIVIDUALS HAPPENS



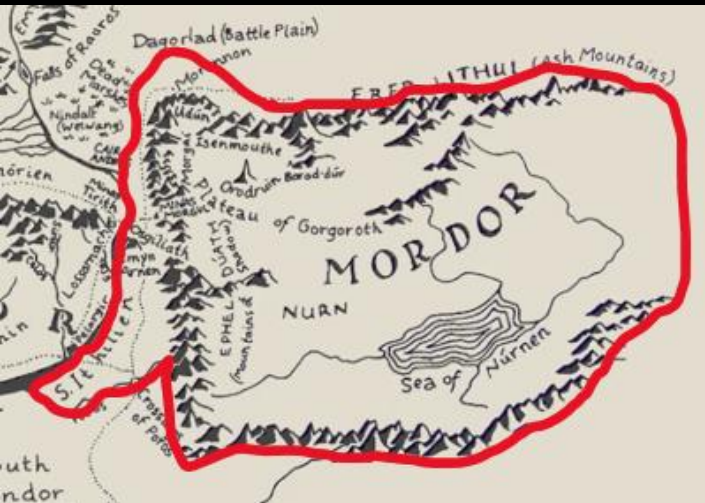
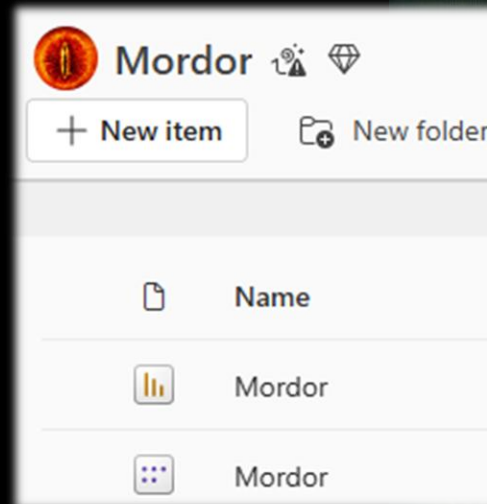
The image is a composite. On the left, there is a vertical list of names, each preceded by a small icon. The icons are: a document icon for 'Name', a bar chart for 'Bree', a grid of dots for 'Bree-hill', a bar chart for 'Chetwood', a grid of dots for 'Hobbit_Simple', and a bar chart for 'Chetwood'. On the right, there is a photograph of a landscape. It shows a river in the foreground, a hobbit house with a lit window in the middle ground, and a grassy hill in the background. The sky is blue with some clouds.

MORDOR

EXTREMELY STRONG ORGANISATION

UNIFORM LOOK AND FEEL

LOW INNOVATION



MISTY MOUNTAINS GOBLINS

NO ORGANISATION

NO AGREEMENTS

NO UNIFORMITY

FREE FOR ALL

Workspaces

Search

My workspace

All

agh morg dirath

Big one

dqfzcazd

Gurshuk

I hate dwarves and Thorin specifically

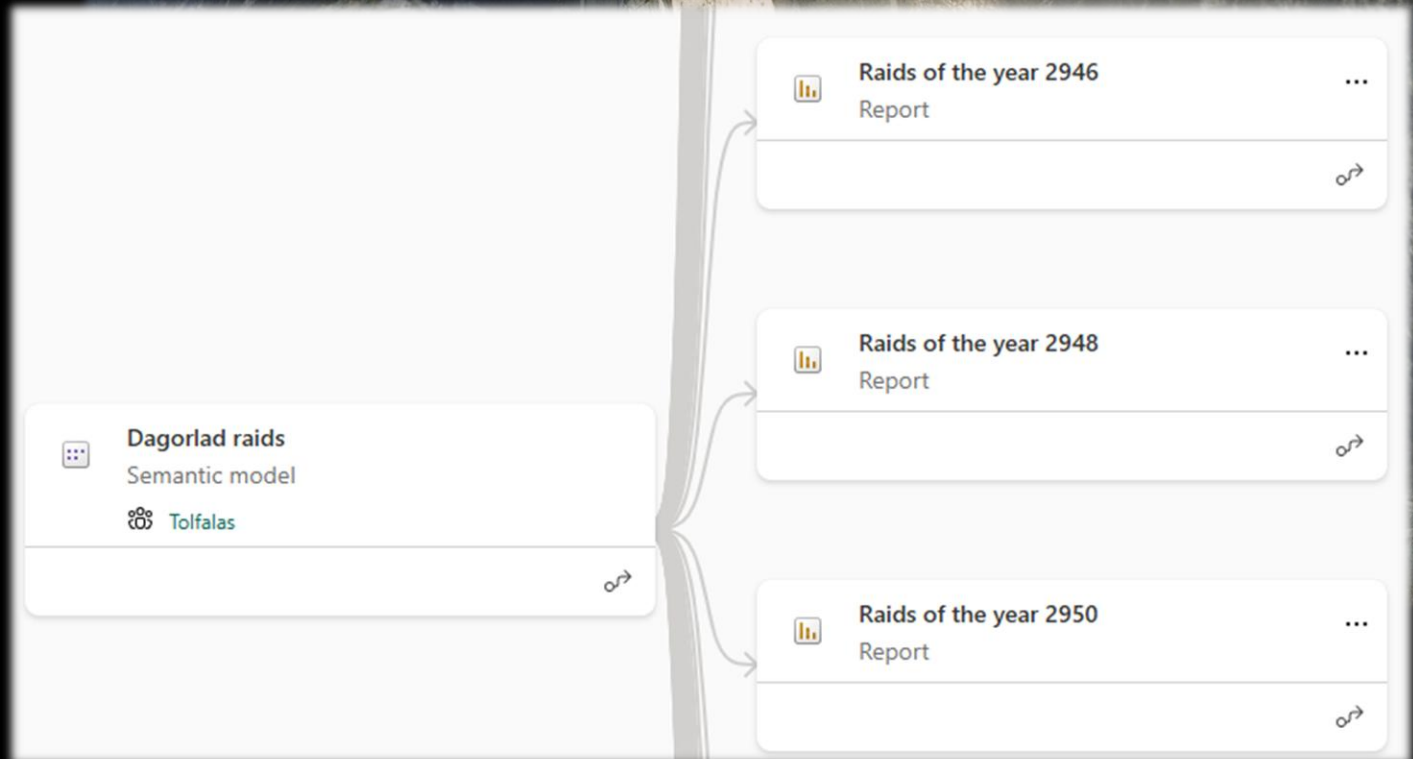


GONDOR

STRONG ORGANISATION

LITTLE TO NO DIFFERENCES

UNIFORM LOOK AND FEEL



TEAMS OF MIDDLE EARTH

CENTRALISED

COMMON THEME

DEPARTEMENTALISED

AGREEMENTS

MANY MEASURES

DECENTRALISED

INNOVATION

TRADITION

NO MEASURES

BAD RELATIONSHIPS

GOOD SEMANTIC MODELS

ADMIN



- HOW TO GET AN OVERVIEW
- WHAT IS FABRIC
- PYTHON AND NOTEBOOKS
- WHAT IS SEMANTIC LINK
- WITHOUT TENANT POWERS

THE YELLOW WIZARD

MIDDLE EARTH & US



STORY OF THE YELLOW WIZARD

REAL-LIFE COMMENTARY BY ME

Before Coffee



JO DAVID – MY JOURNEY

C#
SSRS
TSQL



- DEVELOPMENT
- TRAINING
- COACHING
- GOVERNANCE
- ...



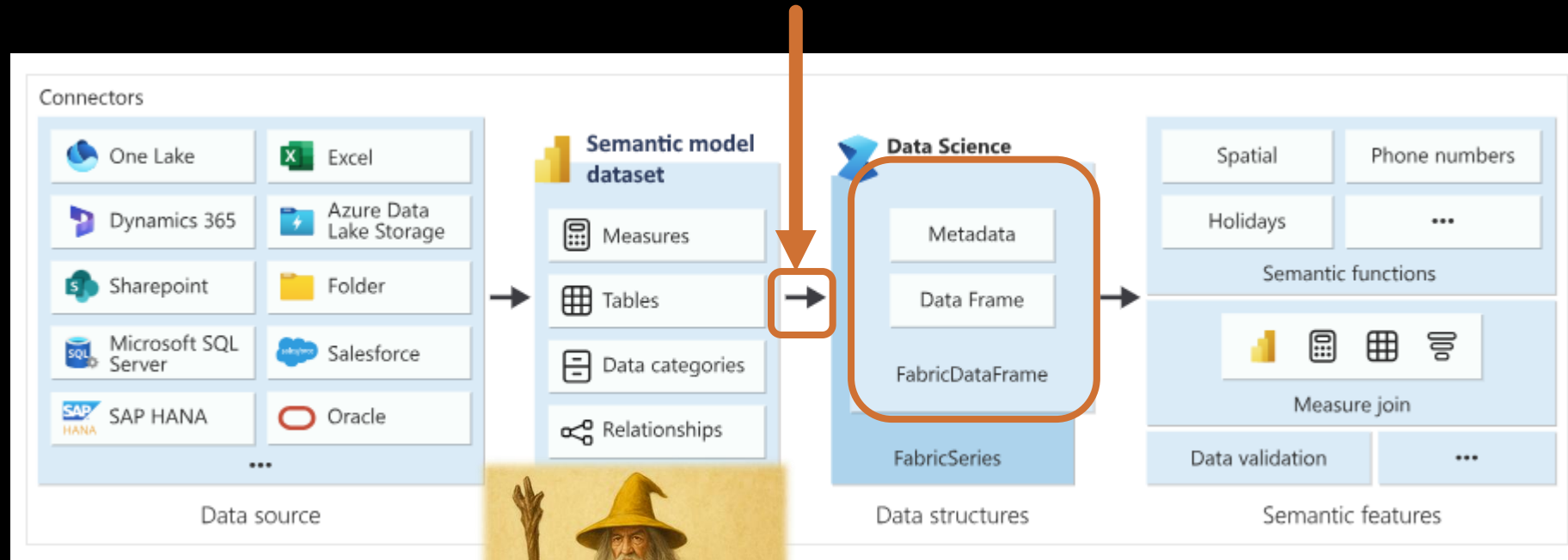


WHAT IS SEMANTIC LINK

- PYTHON SDK
- EXPOSES THE SEMANTIC MODEL AS PYTHON OBJECTS
- CAN READ AND MODIFY MODEL META
- USED IN FABRIC NOTEBOOKS



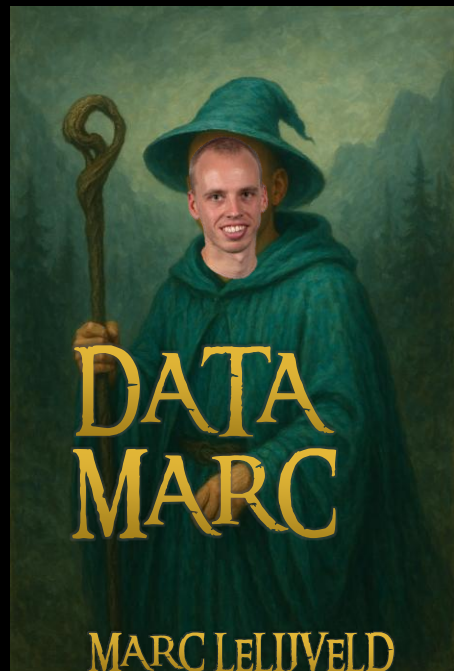
WHAT IS SEMANTIC LINK?



The bridge in the gap between Power BI & the synapse data science experience

FabricDataFrame is the data structure it uses

USAGE OF SEMANTIC LINK

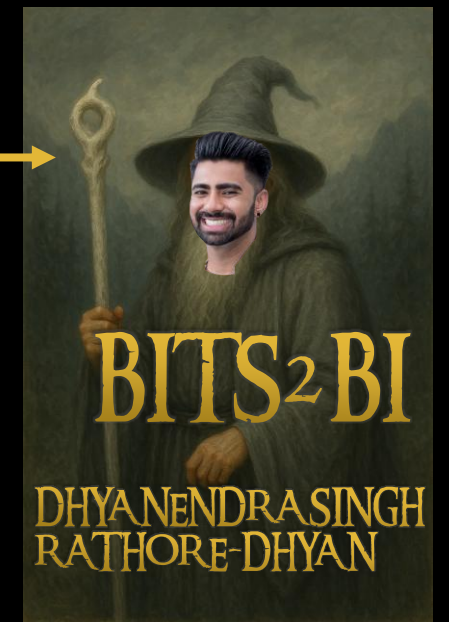


G SEMANTIC MODELS

S

- GOVERNANCE

- TENANT LEVEL
- CAPACITY LEVEL
- BUSINESS LEVEL



PREREQUISITS

`mssparkutils.session.stop()`



InvalidHttpRequestToLivy: [TooManyRequestsForCapacity] This spark job can't be run because you have hit a spark compute or API rate limit. To run this spark job, cancel an active Spark job through the Monitoring hub, choose a larger capacity SKU, or try again later. HTTP status code: 430 {Learn more} HTTP status code: 430.

- A MICROSOFT FABRIC SUBSCRIPTION
- SIGN IN TO MICROSOFT FABRIC
- CREATE A NEW NOTEBOOK OR A NEW SPARK JOB TO USE THIS PACKAGE

WHAT ARE NOTEBOOKS (IN FABRIC)

- WEB-BASED
- CODE FIRST
- DIFFERENT LANGUAGES (PYSPARK, SPARK, SPARK SQL AND R)
- NATIVE INTEGRATION WITH LAKEHOUSES



WHAT ARE NOTEBOOKS (IN FABRIC)

```
1 first_parameter = 5
2 second_parameter = 3
```

✓ 15 sec - Session ready in 14 sec 755 ms. Command executed in 476 ms by jodavid on 10:48:26 PM, 10/27/25

```
1 third_parameter = first_parameter * (second_parameter + first_parameter)
```

✓ <1 sec - Command executed in 389 ms by jodavid on 10:48:27 PM, 10/27/25

```
1 fourth_parameter= third_parameter - first_parameter
```

✓ <1 sec - Command executed in 348 ms by jodavid on 10:48:27 PM, 10/27/25

```
1 display(f"{first_parameter}, {second_parameter}, {third_parameter}, {fourth_parameter}")
```

✓ 2 sec - Command executed in 2 sec 767 ms by jodavid on 10:48:56 PM, 10/27/25

'5, 3, 40, 35'

Code is in cell based
code blocks.
This is easy to debug or
test things



WHAT ARE DATAFRAMES

```
1 import pandas as pd
2
3 df = pd.DataFrame({
4     'Kingdom': ['Gondor', 'Rohan'],
5     'King': ['Aragorn', 'Theoden'],
6     'Capital': ['Minas Tirith', 'Edoras']
7 })
8 display(df)
```

	ABC Kingdom	ABC King	ABC Capital
1	Gondor	Aragorn	Minas Tirith
2	Rohan	Theoden	Edoras

A dataframe is a data structure constructed with rows and columns, similar to a database or Excel spreadsheet.



GETTING STARTED



GREAT, WHAT ARE WE INSTALLING

- SEMANTIC LINK?
 - SEMPY?
- SEMANTIC LINK LABS?

FOLLOW YOUR NOSE (AKA: CHOOSE)

- M*
l
c
r
o
s
o
f
t
- SEMANTIC LINK
 - META-PACKAGE THAT DEPENDS ON THE INDIVIDUAL PARTS
 - SEMPY (OR FUNCTIONS-HOLIDAYS OR GEOPANDAS OR....)
 - PACKAGES THAT CONTAIN CORE FUNCTIONALITY
 - => THE INDIVIDUAL PARTS

- SEMANTIC LINK LABS
 - EXTENDS CAPABILITIES OF SEMANTIC LINK

Community

THE START

SEMANTIC LINK

- INSTALLED IN FABRIC BY DEFAULT
- IMPORT IT TO START

```
1 import sempy.fabric as fabric
```

SEMANTIC LINK LABS

- NEEDS TO BE INSTALLED

```
1 %pip install semantic-link-labs
```

- OR CREATE AN ENVIRONMENT

Environment



Set up shared libraries, Spark compute settings, and resources for notebooks and Spark job definitions.



THE FIRST



Get the list of capacities

```
1 capacities = fabric.list_capacities()
```

```
1 display(capacities)
```

Work with the result in the
second block

By splitting the code, we
can work with the list
without fetching it each
time



DEMO

INSTALLATION

GET DATA

CODE BLOCKS

ITERATING

```
1 workspaces = fabric.list_workspaces()
2 sparkdf = spark.createDataFrame(workspaces)
```

```
1 for _, row in workspaces.iterrows():
2     Id = row["Id"]
3     temp_items = fabric.list_items(workspace=Id)
4     itemdf = pd.DataFrame(temp_items)
```



THE WRITING

```
1 workspaces = fabric.list capacities()  
2 workspaces = fnc_PrepareColumns(workspa  
3 sparkdf = spark.createDataFrame(workspa  
4 sparkdf.write.format("delta").option("m
```

Explorer

Add lakehouses

- ✓ LH_SemanticLink_Data
 - ✓ Tables
 - > Landing_Fabric_Capacities
 - > Landing_Fabric_Columns
 - > Landing_Fabric_Items

Overwrite <> append

```
1 def fnc_PrepareColumns( _Columns):  
2     _Columns.columns = _Columns.column  
3     _Columns.columns = _Columns.column  
4     return _Columns
```

```
4 sparkdf.write.format("delta").option("mergeSchema", "true").option("overwrite", "true").write(f"{lh_abfs_path}/Tables/{Table_Name}")
```



Save to table after
creating a dataframe



THE BEACONS ARE LIT

AKA: LIST OF USEFUL THINGS

- `LIST_CAPACITIES()`
- `LIST_WORKSPACES()`
- `LIST_ITEMS(WORKSPACE)`
- `LIST_MEASURES(WORKSPACE, DATASET)`
- `LIST_RELATIONSHIPS(WORKSPACE, DATASET)`
- `LIST_WORKSPACE_USERS(WORKSPACE)`
- `LIST_COLUMNS(WORKSPACE, DATASET, EXTENDED)`



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DEMO

WRITING
ITERATING

MODELLING



STAR schema ALL THE THINGS

MODELLING

Workspaces

COLUMNS

Users

SEMANTIC LINK

MEASURES

RELATIONS



DEMO

MODELLING
& REPORTING

LAST WORDS OF IMPORTANCE

you are not alone

I CAN'T CARRY IT FOR YOU... BUT I CAN CARRY YOU

ALLOW YOURSELF TO BE CARRIED

By THOSE WHO ARE TRAVELING THE SAME ROAD:



MARC LEIVELD



DHYANEENDRASINGH
RATHORE-DHYAN



prathy.com

sqlbi.com

fabricguru.com



[HTTPS://GITHUB.COM/DATAMINSTREL/DATAMINSTREL_REPO/TREE/MAIN](https://github.com/dataminstrel/DataMinstrel Repo/tree/main)

Thank you for your attention



May SEMANTIC LINK BE THE LIGHT IN DARK PLACES,
when ALL OTHER LIGHTS GO OUT

THE REVEAL



